MEHZABEEN AZAD CHOWDHURY

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ACADEMIC & RESEARCH PROFILE

Computer Science and Engineering graduate pursuing an MSc in Data Science, with nearly three years of professional experience in SaaS product management, software quality assurance, and cross-functional collaboration. Bringing practical experience in requirements definition, software validation, and product workflows into applied healthcare research. Current work includes dental radiographic dataset and annotation quality checks, a collaborative breast cancer classification project, and an accepted MSc thesis proposal on ovarian cancer biomarkers. Interested in clinically meaningful, reproducible, and interpretable computational approaches.

RESEARCH INTERESTS

Clinical and biomedical informatics • Applied medical AI • Clinical biomarker analysis • Diagnostic decision support • Medical imaging data • Explainable ML • Healthcare technology design and evaluation

EDUCATION

Shahjalal University of Science and Technology (SUST), Bangladesh | *MSc in Data Science* | *In progress, 2026–present*

Relevant coursework: Probability & Statistics, Advanced Machine Learning & AI, Advanced Python, High-Dimensional Data Visualization, Mathematics for Data Science, Advanced Data Structures & Algorithms.

Leading University, Sylhet, Bangladesh | *BSc in Computer Science and Engineering* | *2020–2024*

CGPA: 3.44/4.00. Academic preparation in software development, data-oriented computing, software engineering, and quality assurance.

RESEARCH & APPLIED PROJECTS

Ovarian Cancer Diagnosis Using Blood-Based Biomarkers | MSc thesis proposal accepted; research preparation | Supervisor: Dr. Nahid Sultana

- Research direction: investigate benign-versus-malignant ovarian tumor classification using blood-based biomarkers, including CA125, HE4, and selected routine laboratory measures, subject to dataset suitability.
- Planned work includes literature review, dataset suitability assessment, preprocessing, comparison of established classification methods, and evaluation of model performance, calibration, interpretability, and clinical relevance. Research implementation has not yet begun and no results are claimed.

A Multimodal Dental Dataset for Predictive and Assistive Dentistry AI Models | Co-author, manuscript in preparation | Research supervision: Md Saidur Rahman Kohinoor

- Co-author on an interdisciplinary dental imaging dataset manuscript describing 849 periapical radiographs and 301 panoramic radiographs, with a 21-class framework for anatomical structures, dental pathologies, and clinical status.
- Contributing to image and annotation data review, dataset curation, and image–annotation consistency checks, including identification of missing or unmatched entries. Supporting technical validation and documentation in collaboration with the research team and dental experts using CVAT and Google Colab.

Interpretable Breast Cancer Classification | Three-person collaborative applied ML project | In progress

- Using the UCI Wisconsin Diagnostic Breast Cancer dataset to investigate benign and malignant classification, interpretable baseline models, and comparative evaluation of model behavior.
- An initial web landing-page prototype has been developed; model analysis and prediction explanation workflows are in development. A related research proposal is under review. This is an educational research demonstration, not a clinically validated diagnostic tool.

PROFESSIONAL EXPERIENCE

Agency Handy | Product Manager

- Led product requirements and coordinated SaaS feature delivery across design, engineering, and quality assurance. Translated user needs into specifications, product workflows, and development priorities.
- Worked on user onboarding, feature documentation, release coordination, product analytics, and feedback-driven improvements. Applied requirements analysis, validation, and cross-functional communication to practical software delivery.

Technext | Junior Product Management & Software QA

- Contributed to SaaS product development through feature testing, issue identification and documentation, release coordination, and collaboration between product and engineering teams. Evaluated software behavior against requirements and supported iterative improvements.

Appifylab | Product Coordination, QA & Technical Support

- Supported product coordination, functional testing, technical troubleshooting, and communication of user-reported issues to engineering teams. Helped clarify problems, document findings, and coordinate responses across product and support functions.

TECHNICAL, RESEARCH & COLLABORATION SKILLS

Data & research: Python, tabular data cleaning and inspection, XML/image annotation review, exploratory analysis, introductory statistical and ML workflows; pandas, NumPy, and scikit-learn (developing proficiency).

Software & collaboration: Jira, Trello, Notion, Figma, Git/GitHub (developing workflow), PostHog, Postman; product requirements, software QA, user acceptance testing, technical documentation, and interdisciplinary teamwork.

RESEARCH DEVELOPMENT

Current priorities include strengthening reproducible Python analysis, reading and synthesizing clinical AI literature, assessing dataset suitability, and presenting model behavior and limitations to nontechnical healthcare audiences.